Matrix completion in network analysis

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Joint work with

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Network analysis

- "Traditional" multivariate data: properties of individual entities (features, covariates), $n \times p$ data matrix
- Network data is observed relational data: relations between entities, $n \times n$ adjacency matrix
- Can observe relations and "traditional" features at the same time

Examples

- Social media: nodes = users, edges = friends/followers/etc
- Ecology: nodes = animal species; edges = predator/prey
- Biology: nodes = genes and proteins, edges = regulation
- Neuroscience: nodes = brain locations, edges = connections
- Internet traffic: nodes = servers, edges = packages sent
- Transportation: nodes = airports, edges = direct flights
- Political science: nodes = countries, edges = trade, conflicts, etc

Vocabulary and notation

- Network: a set of n entities (nodes) and observed relations between them (edges)
- Represented by an $n \times n$ adjacency matrix A:

$$A_{ij} = \mathbf{1}(\text{there is an edge from } i \text{ to } j)$$

- Undirected network (e.g., i and j are friends): $A = A^T$.
- Directed network (e.g., i sent j an email): A does not have to be symmetric.
- Self-loop: an edge from a node to itself ($A_{ii} = 1$), impossible in some contexts, but in this talk we won't worry about them
- More generally, an edge can have a weight and/or a sign

Probabilistic models for networks

- A_{ij} can be modelled as Bernoulli, Poisson, etc
- Independent edges assumption: not always realistic, but vastly simplifies analysis
- $P = EA = [P_{ij}]$: this $n \times n$ matrix contains all the information
- Structural assumptions on P have to be made: e.g., P is block-constant (communities), low-rank, smooth after a permutation
- Many models are now available, often with many methods for fitting each model
- Many questions: community detection, link prediction/denoising, modeling over time, etc

The general principle

Network analysis methods work when

1. The information of interest can be recovered from EA
2. The probability model is such that A concentrates around EA, i.e. $\|A - EA\|$ is not too large

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- If the structure of interest is in the eigenvectors of EA, we need concentration in matrix spectral norm, plus Davis-Kahan theorem
- Fairly general concentration results available: see e.g. review paper *Le, Levina & Vershynin (2018)*

Cross-validation on networks

- How do we choose among the many models, or set tuning parameters?
- Cross-validation or data splitting: a universal strategy
- How do we split or resample a network in a way that preserves its structure?

Edge cross-validation (ECV)

Li, Levina, Zhu (2018+)

- NCV *(Chen and Lei, 2018)* samples rows and reconstructs the matrix from them; works for block models
- Our goal: a general method not dependent on blocks
- Assume independent edges and write $EA = P = [P_{ij}]$.
- Key assumption: P is low-rank, often observed in networks
- Works for both directed and undirected networks, and with edge weights

The ECV algorithm

Repeat for N random splits (or folds):

1. Split node pairs at random into training and test sets
2. Recover a full matrix $\widehat{A}$ from the training pairs by matrix completion
3. Fit the model on $\widehat{A}$
4. Compute the loss function of interest on test node pairs

Pick the model or parameter to minimize average loss

Matrix completion

Candès and Recht, 2008; Candès and Tao, 2010; Candès and Plan, 2010; Davenport and Plan, 2014; Chatterjee, 2012

- Extensively studied, mostly from applied math and compression perspectives
- General formulation: a matrix A is incompletely observed, with a known mask, where $M_{ij} = 1$ if A_{ij} is observed and 0 otherwise
- Goal: full matrix $\hat{A}$ as close to A as possible
- Noiseless matrix completion: require $M \circ \hat{A} = M \circ A$
- Noisy matrix completion: minimize a loss function $\ell(M \circ \hat{A}, M \circ A)$
- 1-bit matrix completion: both A and $\hat{A}$ are binary

Solving the matrix completion problem

- Assumptions/constraints are necessary
- Low rank $\hat{A}$
- Entries missing completely at random: M_{ij} are i.i.d. Bernoulli
- Research focused on how to best solve the optimization problem

$$\min_{Q \in \mathbb{R}^{n \times n},\ \mathrm{rank}(Q) \leq r} \ell(M \circ A, M \circ Q).$$

- Rank constraint replaced with various convex relaxations

Return to ECV

$$A = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix} \implies \begin{pmatrix} * & 1 & * & 1 \\ 1 & 0 & 1 & 0 \\ * & 1 & 0 & * \\ 1 & 0 & * & 0 \end{pmatrix} \implies \hat{A}$$

- Missing completely at random by construction!
- Makes sense for $P = EA$ low-rank: very common in networks
- Noisy completion: EA is a more useful quantity than A

Why does this work?

- By assumption of independent entries of A, $\hat{A}$ and test pairs are independent

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- By assumption of independent entries of A, $\hat{A}$ and test pairs are independent
- For a low-rank P, if A is not too sparse, $\hat{A}$ has the same concentration order around P as the complete adjacency matrix:

$$\|\hat{A} - P\| \approx \|A - P\|$$

- This makes $\hat{A}$ "stochastically similar" to A

Choice of matrix completion method

- Any reasonable matrix completion method can be used
- One of the most common formulations:

$$\hat{A} = \min_{\mathrm{rank}(Q) \leq K} \sum_{i,j} M_{ij}(A_{ij} - Q_{ij})^2$$

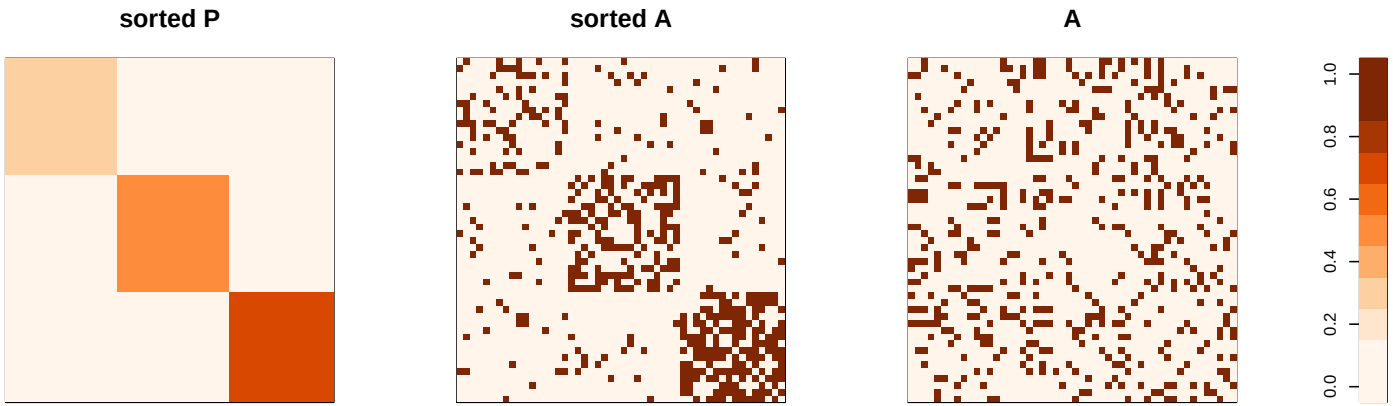
  solved via convex relaxations
- Alternative: singular value thresholding, much faster but slightly less accurate
- An even faster alternative: matrix factorization
- Completion rank: estimate or choose within the same ECV
- Computational cost of ECV depends on both the matrix completion method and the model being fitted

Example: a link prediction algorithm

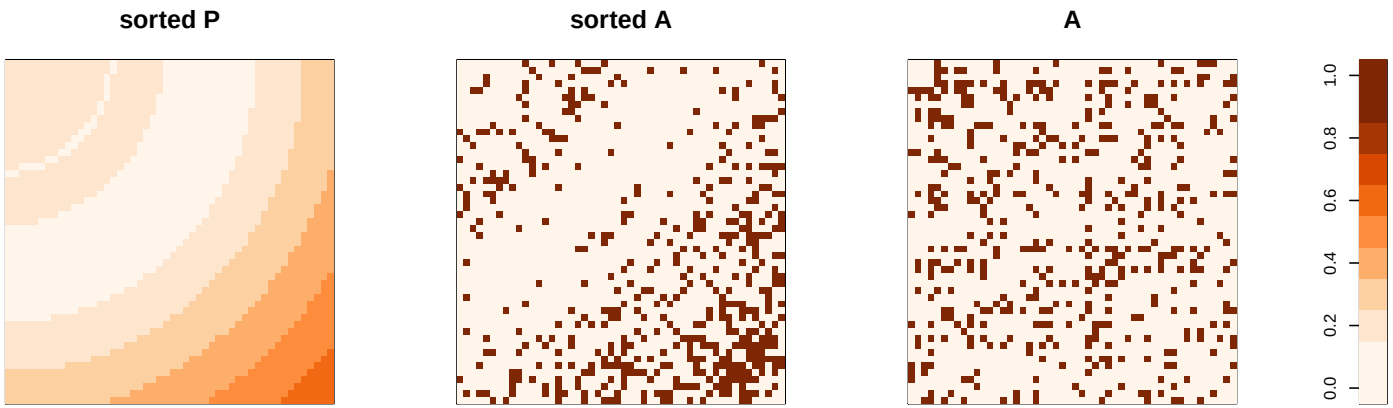
Zhang, Levina, Zhu (2017)

- Uses estimated P to predict/denoise links
- Assumes there is a permutation of nodes that makes P piece-wise smooth
- A neighborhood smoothing algorithm to estimate P from A
- Requires a tuning parameter to control neighborhood size

Synthetic networks

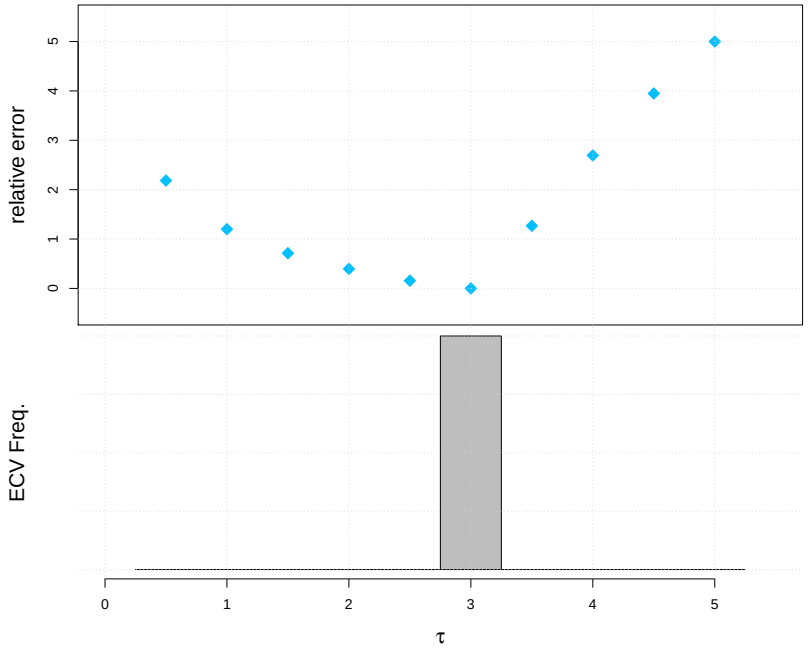


(a) Low rank model

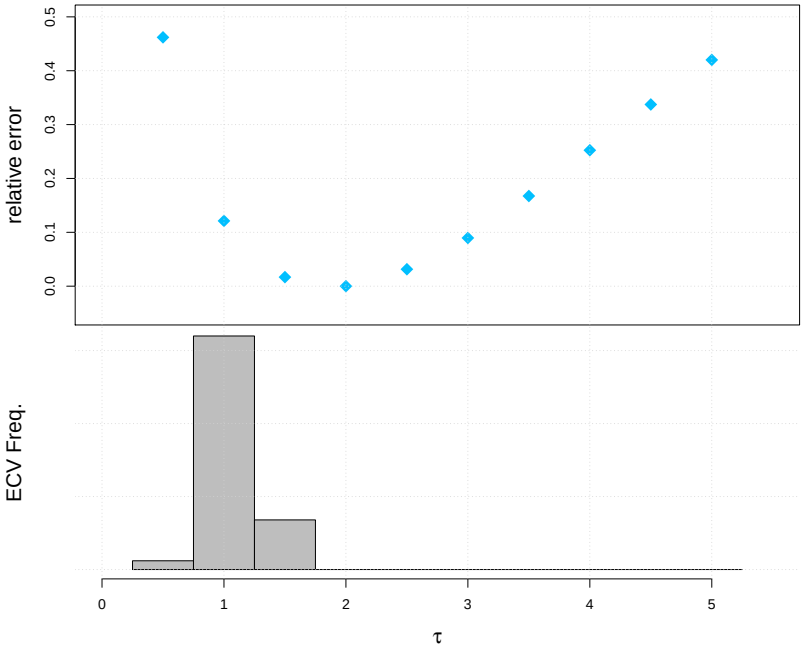


(b) Smooth full rank model

Tuning by ECV



(a) Low-rank model

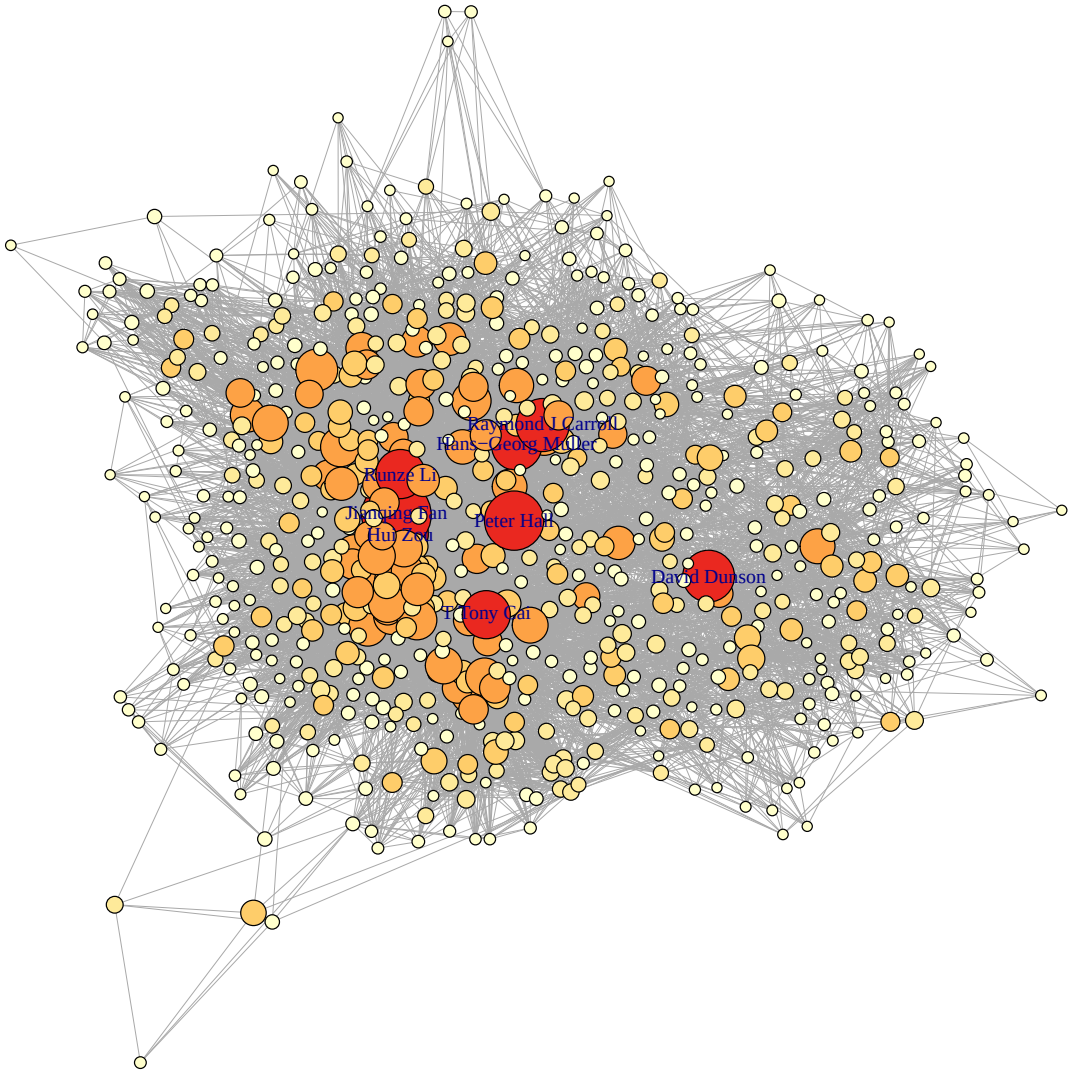


(b) Smooth full rank model

ECV is accurate for a low-rank model but also tunes well when the model is not exactly low rank

Example: community detection

Citation network of statisticians *(Ji and Jin, 2016)*



Regularized spectral clustering

Amini et al (2013)

- Regularize to re-establish concentration: replace A with

$$A_\tau = A + \tau \frac{\bar{d}}{n} \mathbf{1}\mathbf{1}^T$$

  where $\bar{d}$ is average node degree and $\tau \in (0,1)$ a tuning parameter
- Then perform spectral clustering as usual to obtain K blocks
- We can use ECV to choose the values of K and τ
- ECV selects $K = 19$ and $\tau = 1$
- Note this is not assuming a model

Communities of statisticians

Research area (manually labeled)	# communities	# nodes
high-dimensional statistics	7	296
Bayesian methods	3	89
biostatistics	3	83
nonparametrics and semi-parametrics	2	52
functional data analysis	1	52
spatial statistics	1	41
dimensionality reduction	1	39
financial statistics	1	22

Beyond cross-validation

- For ECV, the edges were missing at random by construction
- Many real networks are incompletely observed, but the edges are not likely to be missing at random
- Can matrix completion still help?

Incompletely observed networks

- Many reasons links may be only partially observed
- School friendships

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  - Some kids will name their closest friends first
  - Some kids will name popular or older kids first
  - Typically surveys limit the total
- A hiring network: institution i hired a PhD from institution j

Incompletely observed networks

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- A hiring network: institution i hired a PhD from institution j
  - We see who was hired, but not the offers that were declined
  - Someone was hired but left by the time of data collection
- In many cases the missing mechanism is not independent of the network itself and/or node characteristics

A model-based approach to edge nomination

Li, Levina, Zhu (2018+)

- Goal: incorporate edge nomination mechanism
- Allow edge nominations to depend on node characteristics
- An easy place to start: a model with communities

A directed SBM

- A natural extension of the SBM to directed networks
- Each node i has a community label $c_i \in \{1, \ldots, K\}$
- For all i, j, the variables A_{ij} are independent Bernoulli with

$$P(A_{ij} = 1) = B_{c_i c_j}$$

- The $K \times K$ probability matrix B is not required to be symmetric

Edge nomination mechanism

- Start with directed SBM for A (not observed)
- In nominating friends, each node has its individual preference for communities from which to nominate, determined by how closely they are linked to them, controlled by λ_i
- Each node has individual nominating intensity θ_i
- Let $R_{ij} = \mathbf{1}(A_{ij}$ is reported by node $i)$
- We assume that

$$R_{ij} \sim \text{Bernoulli}(\theta_i B^{\lambda_i}_{c_i c_j})$$

- We then observe $\tilde{A}$ defined by $\tilde{A}_{ij} = A_{ij} R_{ij}$.
- Can directly write

$$\tilde{P}_{ij} = E\tilde{A}_{ij} = \theta_i B^{\lambda_i + 1}_{c_i c_j}$$

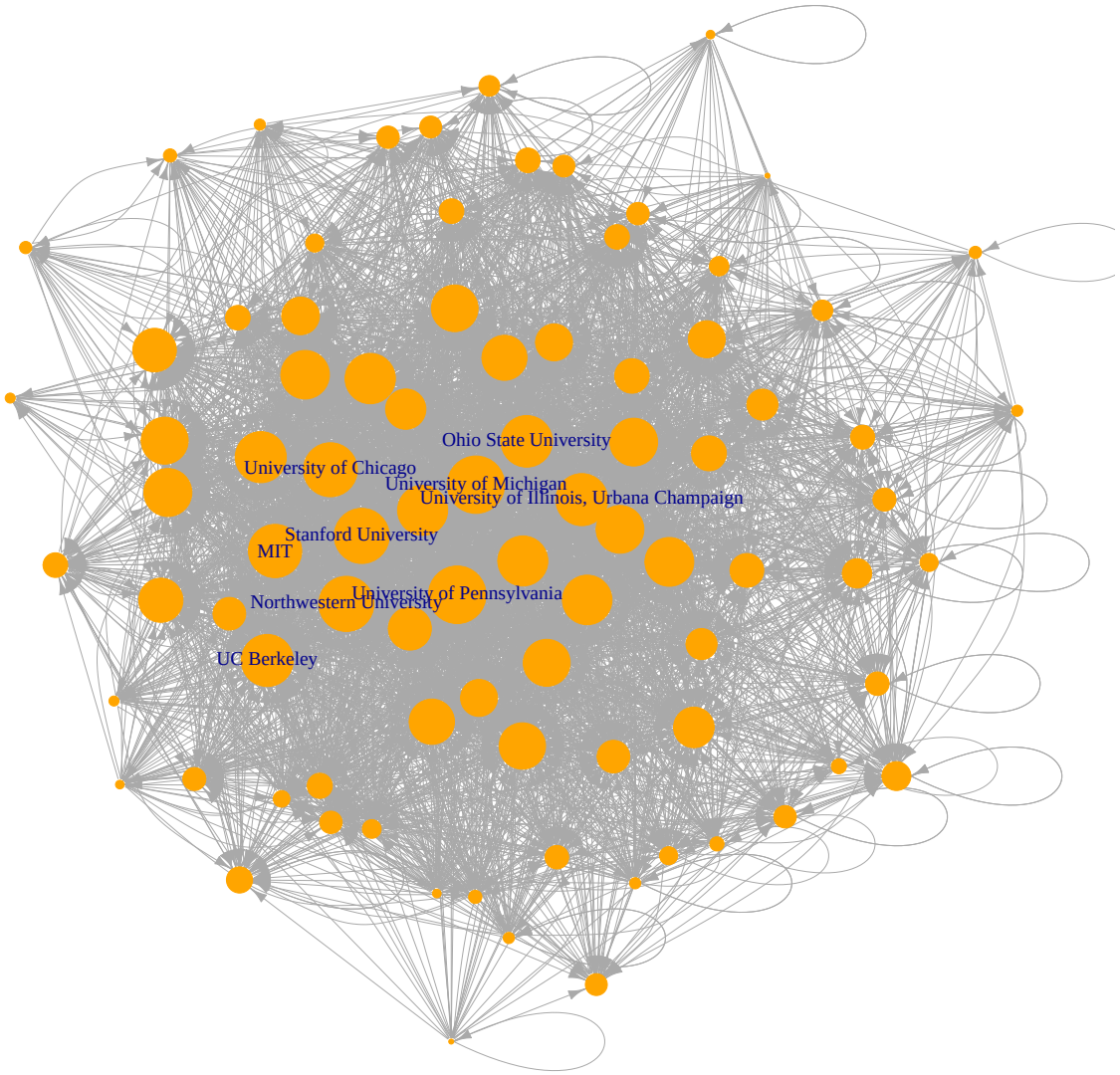
Stochastic block model with nominations (NSBM)

Li, Levina, Zhu (2018+)

- Obtained conditions for identifiability
- Fitting method: method of moments + spectral clustering on the right singular vectors
- Showed consistency
- A conditional version: each node nominates a subset of their neighbors in A, with probabilities modelled in the same way

The hiring network between US business schools

Clauset et al. (2015)



The hiring network between US business schools

- 87 nodes
- Edge $i \to j$: institution i has hired graduates from j
- Edge weight: 1 if i has hired exactly one graduate from j, 2 if more than one
- ECV is used to select the number of communities

Estimated communities

size	US News (avg./med.)	π-ranking (avg./med.)	Top 10 by π-ranking
12	7.7/8	8.3/8	Stanford, MIT, Harvard, UC Berkeley, U. Chicago, Cornell, U. Michigan, Columbia, Yale, U. Penn.
12	29.8/32.5	17.7/17.5	U. Rochester, Northwestern, Carnegie Mellon, U. Wisconsin-Madison, UCLA, U. Minnesota-Twin Cities, UIUC, Purdue, U. Florida
19	53.1/54	45/45	Ohio State, UNC-Chapel Hill, U. Pittsburgh, Penn. State, Indiana U., Michigan State, Georgia Tech, U. Arizona, SUNY-Buffalo, Texas A&M
44	63.7/63	61.4/61.5	Washington U. St.Louis, U. Maryland-College Park, U. Colorado-Boulder, UC Irvine, U. Utah, U. Oregon, U. Southern California, UT-Dallas, U. Virginia, Boston U.

Expected connection weight between communities

$$M_{kl} = \frac{1}{n_k n_l} \sum_{c_i=k, c_j=l} E\tilde{A}_{ij}$$

	Group 1	Group 2	Group 3	Group 4
Group 1	1.86	0.93	0.15	0.06
Group 2	1.56	1.31	0.42	0.13
Group 3	0.86	1.29	0.98	0.32
Group 4	0.76	0.86	0.51	0.22

Shows a clear hierarchy

Individual preference

Institution	$\hat{\lambda}_i$	US News ranking	π-ranking
Yale	2.17	10	11
Cornell	1.88	16	7
Columbia	1.10	9	10
Harvard	0.97	2	3
MIT	0.96	3	2
UC Berkeley	0.85	7	4
U. Chicago	0.75	5	6
Stanford	0.74	1	1
NYU	0.70	10	16
Duke	0.69	12	19
U. Michigan	0.61	14	9
U. Penn	0.57	3	12

Summary

- Edge cross-validation
  - A general computationally feasible cross-validation framework for networks
- Matrix completion
  - Existing methods can sometimes be used directly, as in ECV
  - More complex missing scenarios require custom network models/methods
- Future work
  - Can ECV be adapted to bootstrap?

Thank you

Questions?